

K-FIELD STATISTICAL INWARD POTENTIAL

A Gravity Analogue Generated by Metastable Path Multiplicity

Exact Cell3 propagation, endpoint entropy, anisotropic metric resolution, and the distinction between statistical attraction and mechanical force

Authoritative Cell3 base cell and all 27 centered nodes

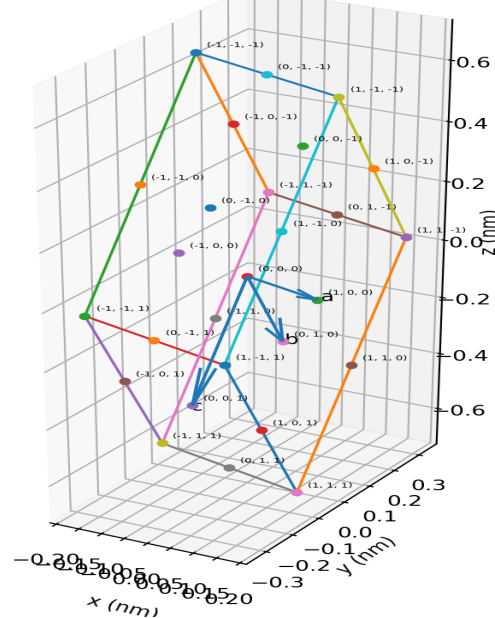


Figure 1. Dimensionally correct Cell3 base cell generated from the authoritative physical basis. All 27 centered nodes are shown and labeled by doubled local address; primitive directions a, b, and c are drawn from the center.

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Abstract

A metastable system that advances through local Cell3 transitions can exhibit an emergent inward statistical tendency even when every microscopic directed edge event is unbiased. The origin of the effect is path-space degeneracy. After a fixed number n of events, nearby endpoints are compatible with many ordered histories containing reversals, loops, and partial cancellations. Distant endpoints require atypically coherent histories in which successive transitions reinforce one another. Summing over histories therefore produces an endpoint probability concentrated near the initial node. This paper defines that concentration as a statistical inward potential, derives its exact 10-event form from all 60,466,176 ordered Cell3 histories, and identifies the associated gravity analogue.

The exact experiment terminates on 891 destination nodes. The starting corner is the most probable single endpoint, with probability 1.939193 percent. The four-edge address shell carries 40.910541 percent of the complete ensemble, whereas the ten-edge ballistic shell carries only 0.760938 percent. The physical endpoint distribution has mean radius 1.685874 nm, RMS radius 1.900285 nm, and maximum available radius 7.923198 nm. Its covariance is anisotropic because the Cell3 primitive edges are unequal and nonorthogonal. The one-standard-deviation principal spreads are 1.651006, 0.725471, and 0.599126 nm.

A dimensionless statistical potential is defined by $\Phi_n(d) = -\ln P_n(d)$ plus an arbitrary constant. Where a physical device assigns an effective energy scale $k T_{\text{eff}}$ to occupancy probability, $U_{\text{stat}} = k T_{\text{eff}} \Phi_n$ and $F_{\text{stat}} = k T_{\text{eff}} \text{grad} \ln P_n$. In the Gaussian regime this becomes an anisotropic linear restoring law, $F_{\text{stat}} = -k T_{\text{eff}} \Sigma_n^{-1} \Delta r$. This is a gravity analogue in endpoint-resolution space: it predicts that the resolved state is more likely to be close than far after n local events. It is not, by itself, a claim that momentum is transferred mechanically. Mechanical force requires an additional coupling that converts occupancy or free-energy gradients into physical work.

Foundational conclusion: Path-space degeneracy creates an effective inward potential in endpoint space. Cell3 geometry fixes the directional metric of that potential; transition weighting changes its scale and anisotropy.

1. Foundations: metastable events and endpoint resolution

A metastable state is a configuration that persists for a finite interval and can transition into one of several neighboring configurations. The present construction uses a rank-three address $d = (d_a, d_b, d_c)$ in the integer lattice. Each event changes exactly one address component by plus or minus one. The six directed transition channels are $+a$, $-a$, $+b$, $-b$, $+c$, and $-c$.

$$d_{(j+1)} = d_j + s_j, \quad s_j \text{ in } \{+/-e_a, +/-e_b, +/-e_c\}$$

$$P(s_j) = 1/6 \text{ for the unbiased null model.}$$

The event law contains no explicit reference to the origin. Nevertheless, a fixed- n endpoint measurement is not uniform over reachable destinations. Every endpoint receives the probabilities of all ordered histories that terminate there. The endpoint probability is therefore a projection of the full history measure onto destination space.

$$P_n(d) = \text{sum over histories } \gamma \text{ ending at } d \text{ of } P(\gamma)$$

$$= \Omega_n(d) / 6^n \text{ in the unbiased case.}$$

$\Omega_n(d)$ is the number of compatible ordered histories. This multiplicity is the source of the inward statistical tendency. A central or locally close destination can be reached through many cancellation-rich histories. A remote destination can be reached only through histories with a large net directional excess.

The word gravity is used here in a specific mathematical sense: an inward gradient of resolved-state probability, or equivalently an outward increase in statistical potential. The analogue concerns statistical resolution after n events. It becomes a mechanical force only when a physical system converts the probability gradient into an energy gradient and permits momentum exchange.

2. Authoritative Cell3 geometry

The corner lattice is an affine copy of $\mathbb{Z}^3$. Physical position is obtained from the authoritative basis B , whose columns are the primitive edge vectors a , b , and c .

$$r(d) = r_0 + B d$$

$$r_0 = (0.1763513501, -0.1186824299, -0.6373114983) \text{ nm}$$

B (nm)	column a	column b	column c
row x	0.327501242070	0.035330775844	-0.010129317627
row y	-0.054897007886	0.207319850441	-0.389787702364
row z	-0.037487502896	-0.547401085392	-0.689734408363

Quantity	Exact value
a	0.3341796791 nm
b	0.5864108904 nm
c	0.7923197650 nm
angle(a,b)	83.933483 deg
angle(a,c)	80.448130 deg
angle(b,c)	50.363232 deg
Cell volume	0.1179148919 nm ³

The direct metric $G = B^T B$ converts an address displacement into a physical squared distance.

$$R(d)^2 = d^T G d = ||B d||^2.$$

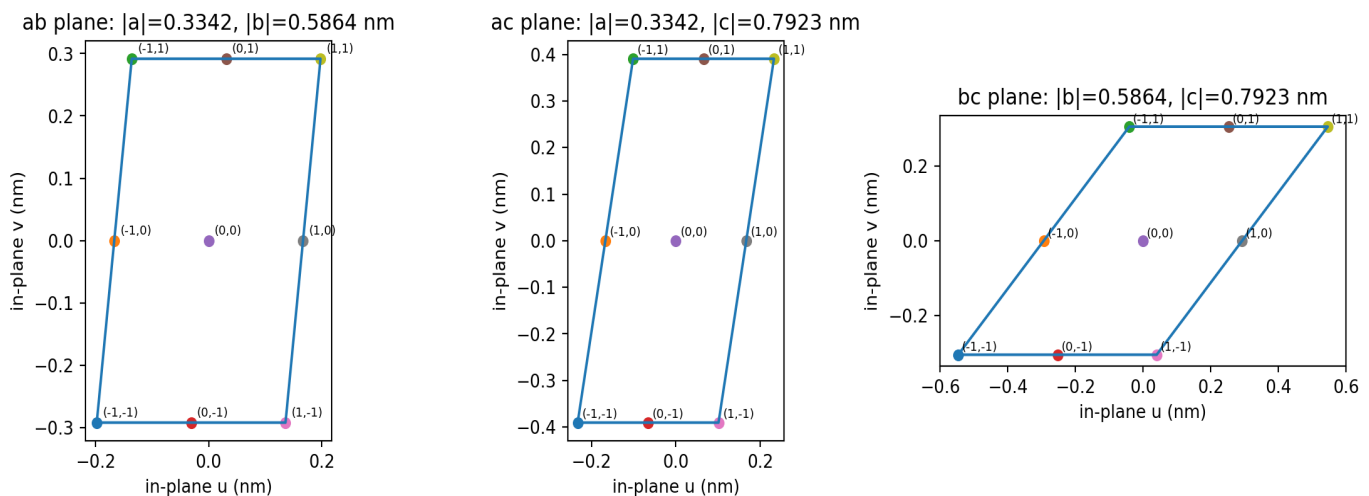


Figure 2. The three Cell3 face planes ab, ac, and bc. Each face is shown in an orthonormal coordinate system intrinsic to that plane, preserving exact physical dimensions and included angles. All nine centered face nodes are labeled.

3. Exact 10-event propagation experiment

Ten events produce exactly $6^{10} = 60,466,176$ ordered histories. Dynamic programming enumerates the Laurent-polynomial coefficient at every endpoint without Monte Carlo approximation.

$$\Omega_{10}(d_a, d_b, d_c) = [x^{d_a} y^{d_b} z^{d_c}] (x + x^{-1} + y + y^{-1} + z + z^{-1})^{10}.$$

The complete history ensemble collapses onto 891 destination addresses. Reachable endpoints satisfy $|d_a| + |d_b| + |d_c|$ in $\{0, 2, 4, 6, 8, 10\}$. The exact shell distribution is shown below.

Minimum edge steps	Destination nodes	Total probability
0	1	1.939193%
2	18	24.428500%
4	66	40.910541%

Minimum edge steps	Destination nodes	Total probability
6	146	25.009734%
8	258	6.951093%
10	402	0.760938%

Exact 891-node endpoint support; size encodes probability

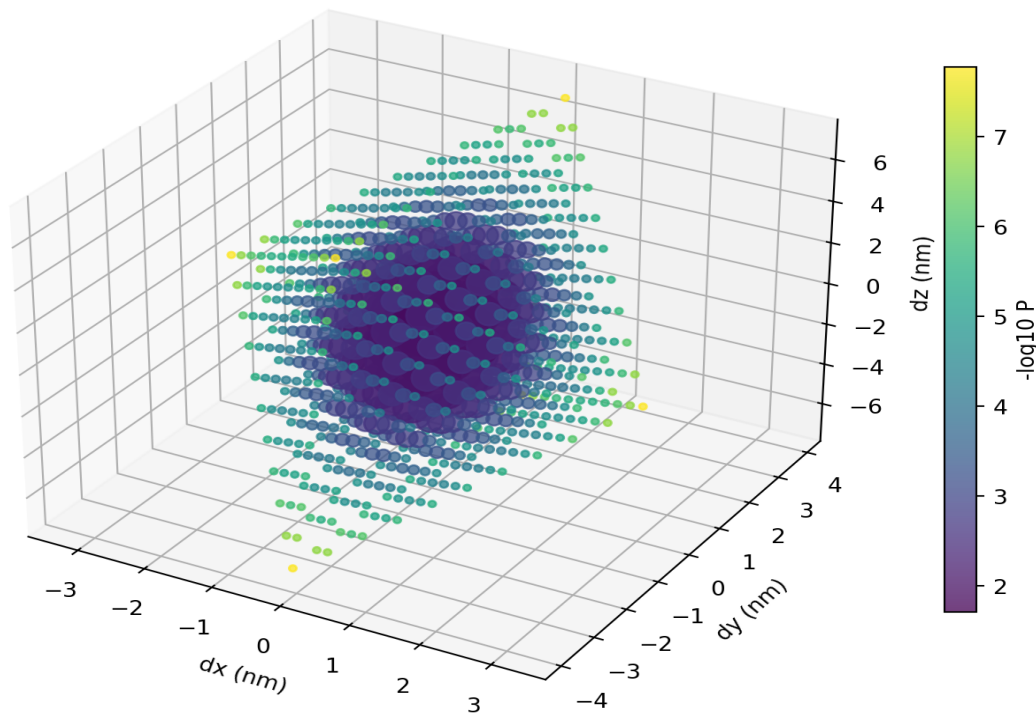


Figure 3. Exact physical positions of all 891 ten-event endpoints relative to the selected starting corner. Marker area increases with endpoint probability; the cloud is centered but strongly anisotropic.

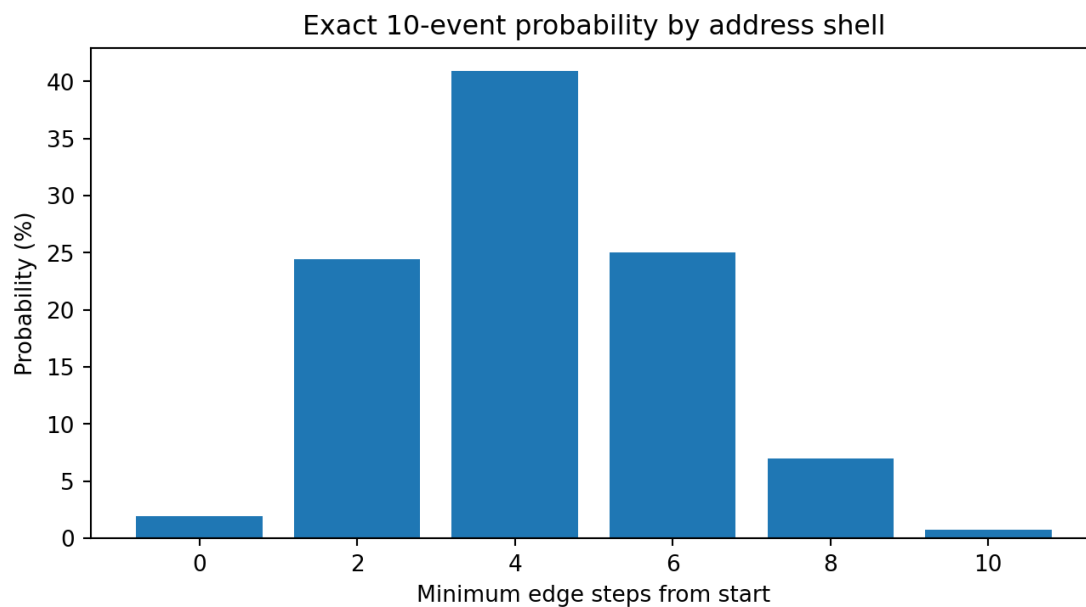


Figure 4. Probability aggregated by minimum edge distance. The four-edge shell, not the outer ten-edge shell, contains the largest probability.

The return multiplicity is $\Omega_{10}(0,0,0) = 1,172,556$, giving return probability 1.939193244 percent. The two extreme endpoints $d = (0,0,\pm 10)$ each have one history and probability $1/60,466,176 = 1.653817\text{e-}8$. The largest distance is $10|c| = 7.923197650$ nm.

4. Why near endpoints dominate

The endpoint is the vector sum of n local transitions. A far endpoint requires a persistent imbalance between positive and negative moves. A near endpoint permits many more combinations because moves may cancel in any order. The effect is combinatorial rather than dynamical: no event must point inward, yet the aggregate endpoint measure is centrally concentrated.

For one coordinate, the number of positive and negative steps must differ by the final displacement. Large displacement fixes a strong imbalance and sharply reduces the number of admissible orderings. In three coordinates, the same restriction acts jointly across the edge families. Each cancellation pair consumes two events while leaving the endpoint unchanged. The center therefore receives histories with the largest freedom to distribute and order cancellation pairs.

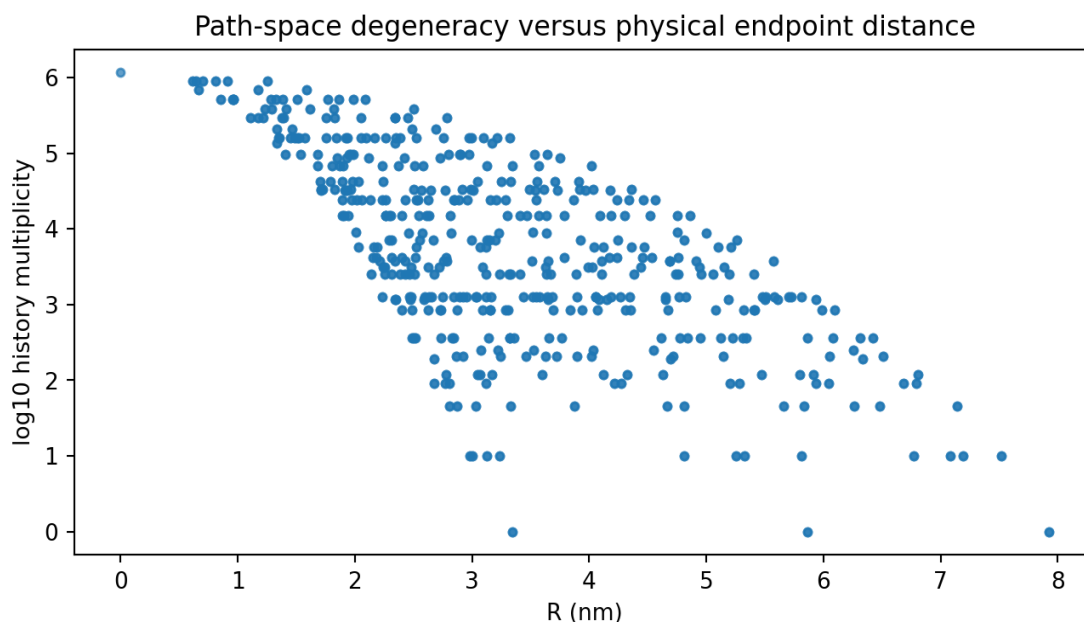


Figure 5. Exact history multiplicity versus physical endpoint distance. Cell3 anisotropy produces scatter at fixed radius, but the upper envelope declines strongly with distance.

The physical radial statistics are exact weighted sums over all endpoints.

Statistic	Distance
Mean	1.685874112 nm
RMS	1.900284540 nm
25th percentile	0.966653867 nm
Median	1.506561770 nm
75th percentile	2.114892459 nm
95th percentile	3.326832199 nm
99th percentile	4.299673100 nm
Maximum	7.923197650 nm

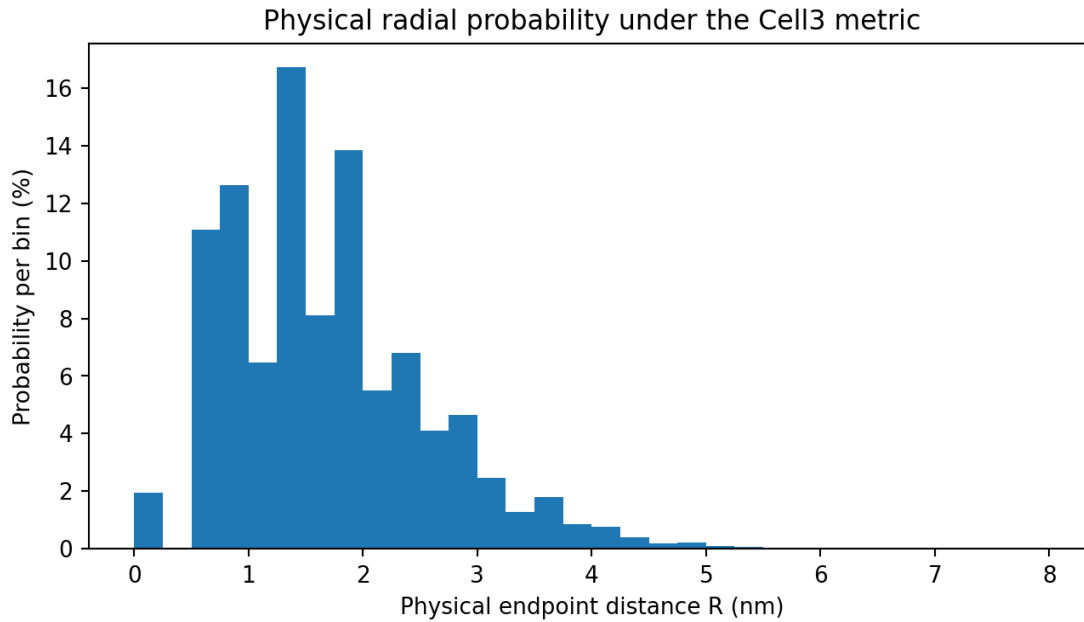


Figure 6. Exact radial probability in physical Cell3 distance. Approximately 76.3 percent of histories terminate within 2.25 nm of the starting corner.

Nearness must be understood statistically. The modal individual endpoint is the origin, while the modal aggregate shell occurs away from the origin because the number of nodes grows with radius. Thus the analogue does not say that every realization returns to the start. It says that endpoint probability decays rapidly relative to the expanding ballistic support and that coherent outward histories are exponentially rare compared with cancellation-rich histories.

5. Statistical potential and force analogue

Any strictly positive probability distribution can be written as an exponential of a dimensionless potential. Define the endpoint statistical potential Φ_n by

$$\Phi_n(d) = -\ln P_n(d) + C_n,$$

where C_n is arbitrary because only potential differences matter. Higher-probability endpoints have lower Φ_n . Since probability generally decreases outward, Φ_n generally rises outward. This is the precise statistical counterpart of an attractive potential well.

If a metastable device has an effective occupancy-energy scale $k T_{\text{eff}}$, define

$$\begin{aligned} U_{\text{stat}}(d;n) &= k T_{\text{eff}} \Phi_n(d) \\ F_{\text{stat}} &= -\text{grad } U_{\text{stat}} = k T_{\text{eff}} \text{grad } \ln P_n. \end{aligned}$$

The force expression is a thermodynamic or information-force relation. Its units are correct only after $k T_{\text{eff}}$ is specified in joules and the gradient is taken with respect to physical distance. Without such an energy scale, $\text{grad } \ln P$ has units of inverse length and describes a statistical tendency rather than a force in newtons.

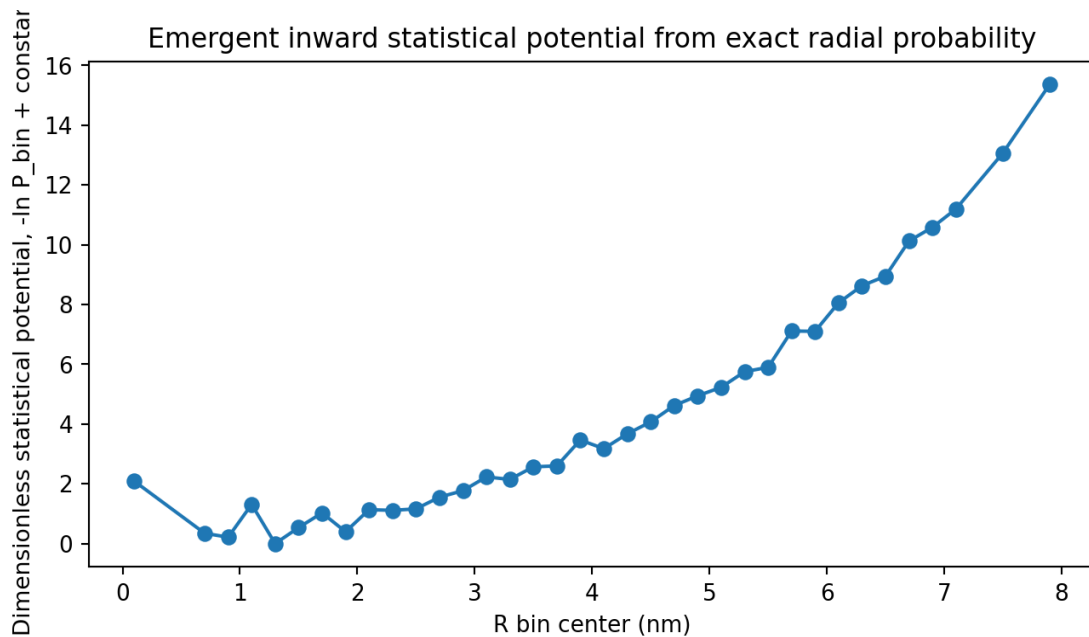


Figure 7. Dimensionless radial statistical potential computed directly from exact radial-bin probabilities. The minimum lies in the high-multiplicity central region and the potential rises toward the ballistic boundary.

For sufficiently large n , the central region is approximately Gaussian.

$P_n(\Delta r)$ approximately $\exp[-(1/2) \Delta r^T \Sigma_n^{-1} \Delta r] / \sqrt{(2\pi)^3 \det \Sigma_n}$
 $\Sigma_n = (n/3) B B^T$ for the unbiased six-edge law.

The corresponding local potential and force are

U_{stat} approximately $(k T_{\text{eff}}/2) \Delta r^T \Sigma_n^{-1} \Delta r$
 F_{stat} approximately $-k T_{\text{eff}} \Sigma_n^{-1} \Delta r$.

This is an anisotropic harmonic gravity analogue. The statistical restoring coefficient is largest along the narrowest covariance direction and smallest along the broadest direction.

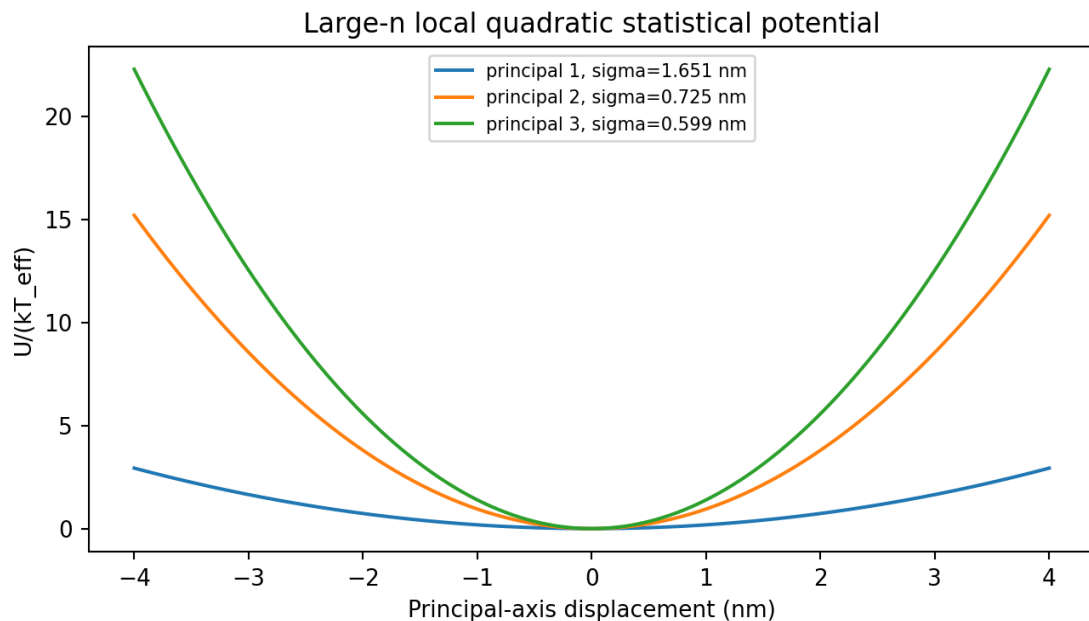


Figure 8. Gaussian-limit potential sections along the three principal covariance axes for $n = 10$. The narrow principal direction produces the steepest dimensionless potential.

6. Large-n limit and scaling laws

The maximum possible displacement grows linearly with event count because all n transitions can align along the longest edge. The RMS endpoint distance grows only as the square root of n because independent signed increments add diffusively.

$$R_{\max}(n) = n \max(|a|, |b|, |c|)$$

$$R_{\text{RMS}}(n) = \sqrt{n/3 * (|a|^2 + |b|^2 + |c|^2)}$$

$$R_{\text{RMS}}/R_{\max} \text{ proportional to } n^{(-1/2)}.$$

The endpoint cloud therefore expands in absolute distance but contracts relative to the full reachable support. This relative contraction is the strongest mathematical basis for the gravity analogy: as more local events occur, the resolved endpoint occupies an increasingly small fraction of the available ballistic radius.

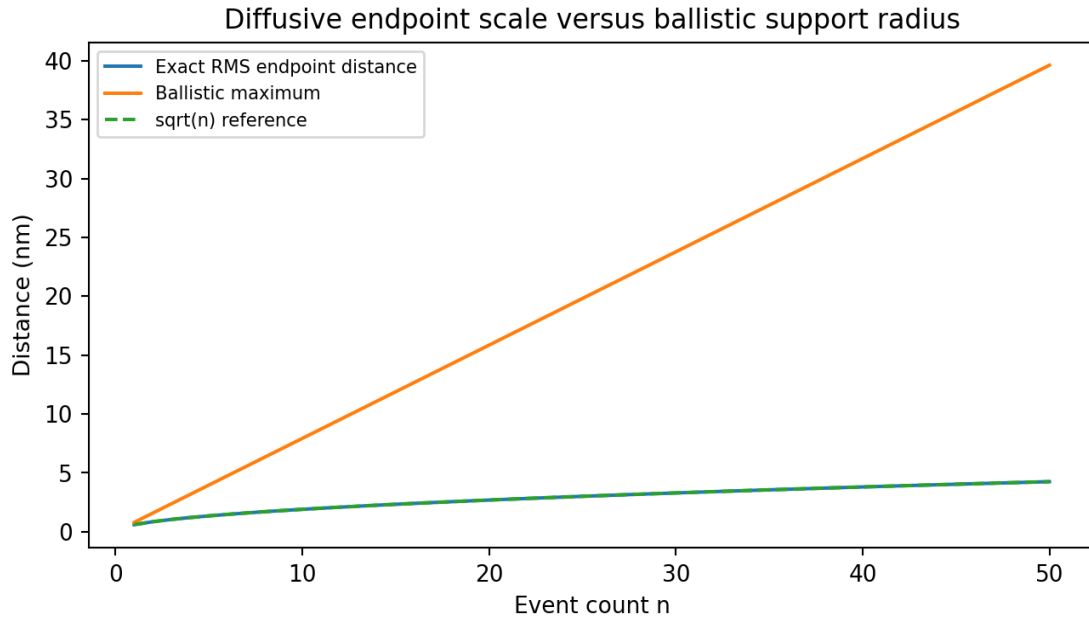


Figure 9. Exact RMS endpoint distance for $n = 1$ through 50 compared with the ballistic maximum and a square-root reference.

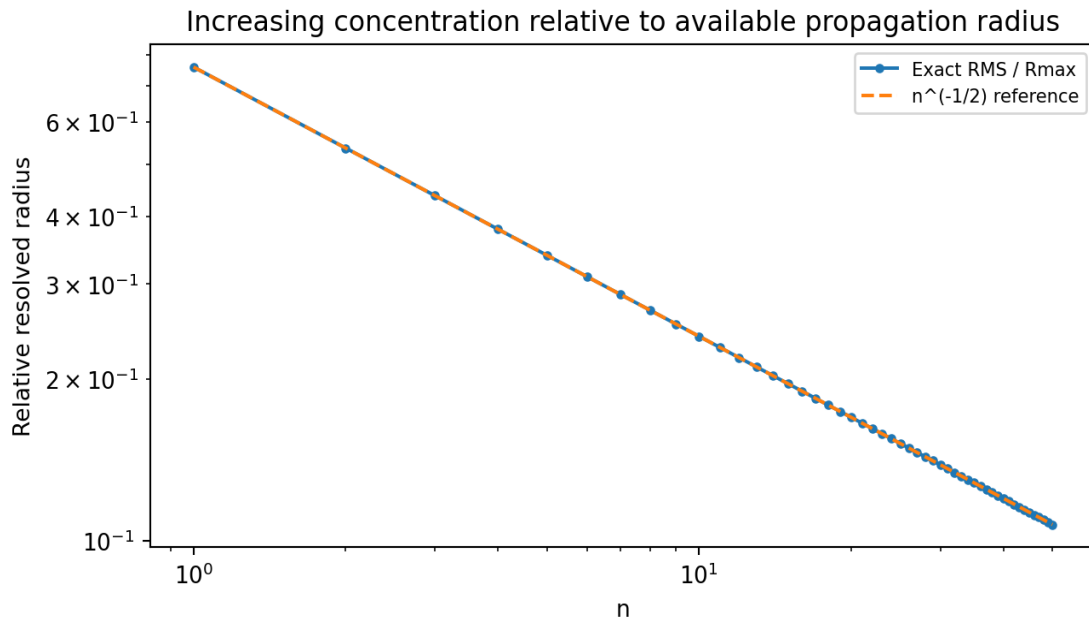


Figure 10. Exact relative concentration $R_{\text{RMS}}/R_{\max}$. The computed sequence follows the $n^{(-1/2)}$ law.

At a fixed physical displacement Δr , the Gaussian statistical force coefficient scales as Σ_n^{-1} proportional to $1/n$. Thus the local gradient at a fixed point weakens as the cloud broadens. At a typical displacement of order $\sqrt{n}$, the typical dimensionless force magnitude scales as $1/\sqrt{n}$. The inward analogue is therefore not a static central field independent of observation time; it is an n -indexed resolution field defined by the event ensemble.

7. Edge-length-weighted metastability

Pure metastability does not uniquely determine how the six channels are weighted. Geometry supplies edge lengths, while the device physics supplies the transition law. Four exact ten-event scenarios are evaluated here: equal directed probabilities; inverse-length family weighting; direct-length family weighting; and a Boltzmann-like weight $\exp(-\ell/\ell_{\text{bar}})$. In every scenario opposite directions within one edge family remain equally likely, so mean displacement remains zero.

$$p_{\text{+}(i+)} = p_{\text{+}(i-)} = w_i / [2(w_a + w_b + w_c)].$$

Scenario	$p_{\text{+}(-a)}$ each	$p_{\text{+}(-b)}$ each	$p_{\text{+}(-c)}$ each	Mean R	RMS R	P(origin)
equal	0.166667	0.166667	0.166667	1.685874	1.900285	1.9392%
inverse length	0.251049	0.143066	0.105886	1.524459	1.695318	2.1856%
direct length	0.097547	0.171174	0.231279	1.811122	2.073387	2.1720%
boltzmann length	0.239102	0.153719	0.107179	1.539590	1.713747	2.1404%

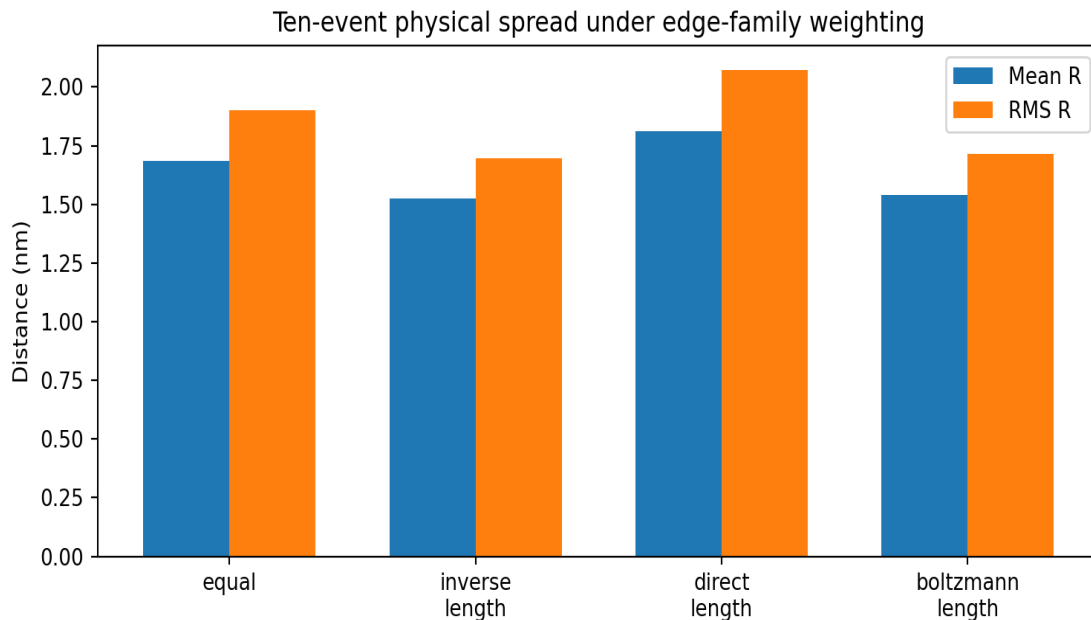


Figure 11. Mean and RMS endpoint distances under four explicit edge-family weighting laws. Short-edge preference reduces physical spread; long-edge preference increases it.

The inward statistical concentration survives all four zero-drift weighting laws because cancellation remains combinatorially favored over ballistic coherence. Weighting changes the metric shape and radial scale. Inverse-length or energy-suppressing weights preferentially use the short a edge and tighten the cloud. Direct-length weighting preferentially uses the long c edge and broadens the cloud. A directional imbalance, such as $p_{\text{+}(c+)} \neq p_{\text{+}(c-)}$, adds drift and displaces the potential minimum away from the initial node.

Accordingly, edge-length weighting and path multiplicity play distinct roles. Path multiplicity generates the central tendency. Edge weighting selects the diffusion tensor and therefore determines how the tendency is expressed in physical space.

8. Anisotropy, reciprocal geometry, and Cell3 planes

The unbiased address walk has covariance $(n/3)I$ in address coordinates, but physical displacement is $B d$. Therefore the physical covariance is $(n/3)BB^T$. Nonorthogonality couples the primitive channels, and unequal edge lengths produce unequal principal spreads.

Principal mode	One-sigma spread	Direction x	Direction y	Direction z
1	1.651006264 nm	-0.035969282	0.246370428	0.968508040
2	0.725470558 nm	-0.008496957	0.969024744	-0.246817435
3	0.599126131 nm	0.999316773	0.017107217	0.032761725

One-standard-deviation Cell3 covariance ellipsoid, $n=10$

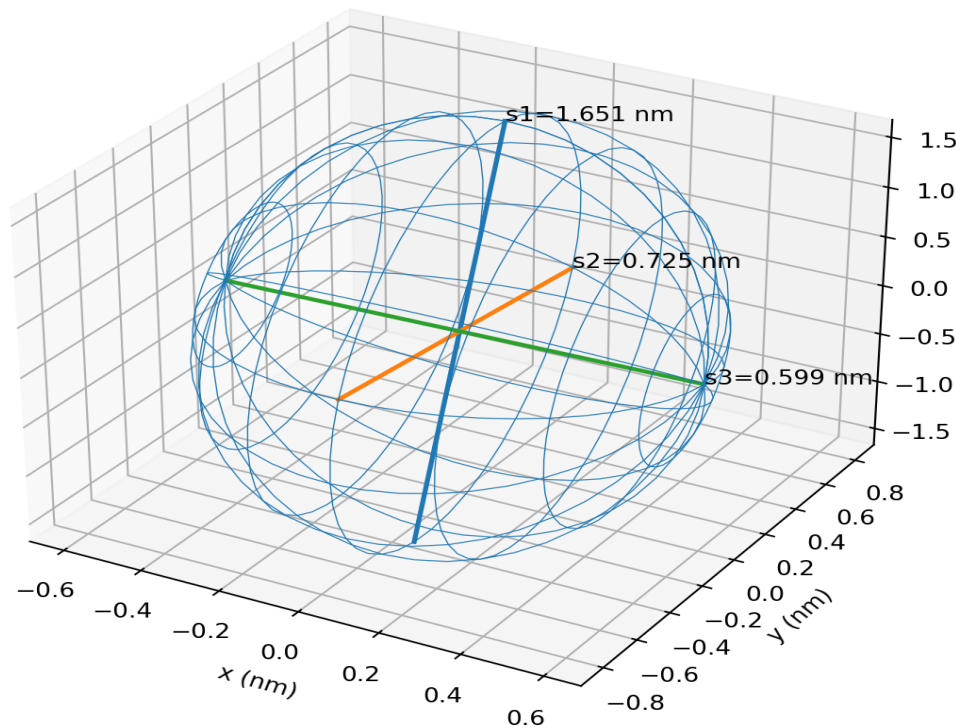


Figure 12. Exact one-standard-deviation covariance ellipsoid. The long axis is dominated by the c-edge orientation and its nonorthogonal coupling to b.

The reciprocal metric $G^{(-1)}$ controls gradients in address coordinates. Because the statistical force is a gradient, its natural directional structure is reciprocal to the direct endpoint covariance. Broad direct-space directions have shallow potential curvature; narrow direct-space directions have steep curvature. This direct-reciprocal pairing is not optional bookkeeping. It is the geometry that converts path multiplicity into a directionally resolved inward tendency.

Direct covariance: $\text{Sigma}_n = (n/3) B B^T$

Reciprocal curvature: $\text{Sigma}_n^{(-1)} = (3/n) B^{(-T)} B^{(-1)}$.

Each Cell3 face plane therefore supports a different two-dimensional projection of the statistical potential. Rotation of a physical device relative to these planes changes how a laboratory coordinate samples the principal curvature axes, even when the underlying event law is unchanged.

9. Physical interpretation and limits

The statistical result is exact: after n local events, near endpoints carry more path measure than endpoints near the ballistic boundary. This is sufficient to define an inward statistical potential. Three interpretive levels must nevertheless remain distinct.

Level 1 - endpoint probability. The probability distribution alone predicts which destination is more likely when the state is resolved. No mechanical force is required.

Level 2 - entropic or free-energy force. If the metastable device is thermally or actively coupled so that occupancy probability corresponds to free energy, the gradient $k T_{\text{eff}} \text{grad} \ln P$ is a physical generalized force. Polymer elasticity and osmotic forces are familiar examples of this class of mechanism, although the Cell3 geometry here is different.

Level 3 - gravitational dynamics. To reproduce conventional gravity, a theory must additionally establish source dependence, mass coupling, momentum conservation, propagation, superposition, equivalence behavior, and the observed distance law. The present result supplies an inward statistical mechanism but does not by itself derive an inverse-square field.

The analogue is locally harmonic in the Gaussian regime, not inverse-square. It is centered on the initial or reset state, and its curvature depends on n . These features make it directly testable and distinguish it from ordinary gravity. They also identify the additional closure required for a full gravitational theory: the event count, reset process, or coherence horizon must become a physical field variable whose spatial dependence transforms the local quadratic statistical potential into the observed macroscopic law.

Exact statement: Pure local metastability statistically favors resolution into close endpoints because close endpoints possess larger path-history fibers. This is a generated inward potential in endpoint space.

10. Experimental observables

A metastable Cell3 device can test the statistical potential without presuming a mechanical gravity interpretation. The primary observable is the complete destination histogram after a fixed number of events. The exact unbiased distribution is the null model. Deviations can be separated into drift, covariance deformation, shell redistribution, and antipodal imbalance.

Observable	Null-model prediction	Directional-coupling signature
Mean displacement	Zero	Nonzero vector proportional to signed edge imbalance
Covariance	$(n/3)BB^T$	Eigenvalue or eigenvector change
Antipodal pair probabilities	Equal	$P(d)$ not equal to $P(-d)$
Return probability	Exact combinatorial value	Excess or deficit of cancellation histories
Shell distribution	Exact table for n	Outward or inward redistribution
Fiber-sensitive variable	May differ at common endpoint	Hysteresis, phase, strain, or polarization conditioned on history

A rotation experiment should hold event count, temperature, drive amplitude, and readout threshold fixed while changing the device orientation relative to the K-Field geometry. A phase-reversal experiment should exchange + and - channel biases and therefore reverse any antipodal imbalance. Repetition at multiple n values tests the predicted $\sqrt{n}$ spread and $1/\sqrt{n}$ relative concentration.

To detect mechanical conversion, the experiment must measure both occupancy statistics and force or work. A correlation between $\text{grad} \ln P$ and a calibrated mechanical response would determine the effective energy scale $k T_{\text{eff}}$ or its nonthermal analogue. Without this paired measurement, a histogram establishes statistical gravity but not force magnitude in newtons.

11. Consequences and new deductions

First, localization and motion are not opposites. The microscopic traversed length grows linearly with n , while net displacement grows only as $\sqrt{n}$. A system may therefore execute extensive internal motion and still resolve close to its initial configuration.

Second, statistical gravity is a property of coarse-graining. The full history ensemble is uniform in the unbiased model, but the endpoint projection is highly nonuniform. Attraction emerges when many histories are identified as one resolved destination.

Third, the endpoint does not contain the complete state. The exact entropy decomposition for $n = 10$ is

```
H_history = 17.917594692281 nats
H_endpoint = 5.364728188272 nats
H_fiber = 12.552866504009 nats
H_fiber/H_history = 70.058882 percent.
```

Approximately 70.06 percent of the history information remains in the fiber after the endpoint is known. Any traversal-order-dependent phase, hysteresis, stored strain, polarization, or coherence can therefore survive endpoint localization and become a secondary observable.

Fourth, the inward potential has a reciprocal-space representation. The direct metric sets displacement cost, while its inverse sets gradient curvature. This means that apparently weak direct-space spreading along one axis corresponds to a steep reciprocal response along the conjugate axis.

Fifth, an inverse-square macroscopic law is not produced by a fixed- n unconstrained walk. Such a law would require a spatially varying event horizon, state-dependent transition rates, coupling between multiple walkers, or a scale-dependent coarse-graining rule. This identifies a concrete research program rather than an ambiguity.

Sixth, a stationary bound distribution requires either resetting, finite boundaries, state-dependent barriers, or dissipation. The free unbiased walk broadens indefinitely in absolute distance. Its gravity analogue is relative concentration within an even faster-growing reachable support, not permanent confinement.

Central theorem. For a symmetric finite-variance local transition law, endpoint probability after n events concentrates on displacements of order $\sqrt{n}$, while the reachable ballistic radius is of order n . Consequently, the probability of resolving at a fixed fraction of the ballistic radius is exponentially suppressed with n .

Appendix A. Exact data tables

A.1 Radial interval probabilities

Physical interval	Probability
R = 0	1.939193%
0 < R ≤ 0.75 nm	11.089340%
0.75 < R ≤ 1.50 nm	35.817611%
1.50 < R ≤ 2.25 nm	27.458310%
2.25 < R ≤ 3.00 nm	15.540225%
3.00 < R ≤ 4.00 nm	6.359116%
4.00 < R ≤ 5.00 nm	1.582997%
5.00 < R ≤ 6.00 nm	0.202133%
6.00 < R ≤ 8.00 nm	0.011074%

A.2 Event-count scaling, selected values

n	Support	Mean R (nm)	RMS R (nm)	Rmax (nm)	RMS/Rmax	Return P
1	6	0.570970	0.600923	0.792320	0.758435	0.00000000
2	19	0.738274	0.849833	1.584640	0.536294	0.16666667
3	44	0.937777	1.040829	2.376959	0.437882	0.00000000
5	146	1.200991	1.343704	3.961599	0.339182	0.00000000
10	891	1.685874	1.900285	7.923198	0.239838	0.01939193
15	2736	2.065454	2.327364	11.884796	0.195827	0.00000000
20	6181	2.381764	2.687408	15.846395	0.169591	0.00710835
30	19871	--	3.291389	23.769593	0.138471	0.00391714
40	45961	--	3.800569	31.692791	0.119919	0.00256003
50	88451	--	4.249165	39.615988	0.107259	0.00183864

A.3 Highest-probability endpoints

d_a	d_b	d_c	Histories	Probability	R (nm)	Fiber entropy
0	0	0	1172556	1.939193%	0.000000	13.974697
-1	-1	0	890820	1.473253%	0.704964	13.699898
-1	0	-1	890820	1.473253%	0.909572	13.699898
-1	0	1	890820	1.473253%	0.807200	13.699898
-1	1	0	890820	1.473253%	0.643532	13.699898
0	-1	-1	890820	1.473253%	1.250773	13.699898
0	-1	1	890820	1.473253%	0.615518	13.699898
0	1	-1	890820	1.473253%	0.615518	13.699898
0	1	1	890820	1.473253%	1.250773	13.699898
1	-1	0	890820	1.473253%	0.643532	13.699898
1	0	-1	890820	1.473253%	0.807200	13.699898
1	0	1	890820	1.473253%	0.909572	13.699898
1	1	0	890820	1.473253%	0.704964	13.699898
-2	0	0	680190	1.124910%	0.668359	13.430127
0	-2	0	680190	1.124910%	1.172822	13.430127
0	0	-2	680190	1.124910%	1.584640	13.430127
0	0	2	680190	1.124910%	1.584640	13.430127
0	2	0	680190	1.124910%	1.172822	13.430127
2	0	0	680190	1.124910%	0.668359	13.430127
-2	-1	-1	509040	0.841859%	1.506562	13.140282
-2	-1	1	509040	0.841859%	0.855955	13.140282
-2	1	-1	509040	0.841859%	0.958371	13.140282
-2	1	1	509040	0.841859%	1.323838	13.140282
-1	-2	-1	509040	0.841859%	1.863128	13.140282
-1	-2	1	509040	0.841859%	0.961433	13.140282
-1	-1	-2	509040	0.841859%	2.090305	13.140282
-1	-1	2	509040	0.841859%	1.283252	13.140282
-1	1	-2	509040	0.841859%	1.383977	13.140282
-1	1	2	509040	0.841859%	1.983693	13.140282
-1	2	-1	509040	0.841859%	0.966654	13.140282

Appendix B. Reproducible exact algorithm

The endpoint ledger is generated by integer dynamic programming. The algorithm stores a map from displacement address to history count. It begins with one history at the origin. Each iteration distributes every existing count to six neighboring addresses. Integer arithmetic remains exact for the complete unbiased calculation.

```
counts = {(0,0,0): 1}
for event in range(n):
    next = empty integer map
    for d, count in counts:
        for s in six directed edge moves:
            next[d+s] += count
    counts = next
    P_n(d) = counts[d] / 6^n
```

Physical coordinates and all derived observables are then deterministic:

```
Delta r = B d
R = sqrt(d^T B^T B d)
Sigma_n = sum_d P_n(d) Delta r Delta r^T
H_endpoint = -sum_d P_n(d) ln P_n(d)
H_fiber = sum_d P_n(d) ln Omega_n(d).
```

The machine-readable companion contains the complete 891-endpoint registry, all exact counts, physical vectors, radii, probabilities, surprisals, fiber entropies, edge-weight scenarios, and $n = 1$ through 50 scaling data.

Appendix C. Archival record

Field	Record
Document title	K-Field Statistical Inward Potential: A Gravity Analogue from Metastable Path Multiplicity
Principal Investigator	P. Dwayne Esterline
Education	BS 1999 Huntington University; MBA 2008 Indiana Wesleyan University
Organization	Krytronx, LLC
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785206671
Output PDF	K-Field_statistical_inward_potential_gravity_analogue_1785206671.pdf
Machine-readable companion	K-Field_statistical_inward_potential_gravity_analogue_1785206671.json
Generation method	Python ReportLab using built-in core fonts only
Authoritative geometry	Physical Cell3 basis B in nanometres; affine corner r0; exact integer-lattice address propagation
Enumeration	Exact dynamic programming; no Monte Carlo sampling
Source continuity	K-Field 10-Event Cell3 Propagation Experiment and established K-Field archival report conventions

End of archival reference.